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BAGONG PILIPINAS

CC 101 – Introduction to Computing

"Know Your Network Settings": Exploring Windows 11 Network Configuration Tools

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Computer Unit Used (Desktop/Laptop, brand): laptop HP

Date: October 20, 2026

I. Objectives

At the end of this activity, you should be able to:

- Locate network adapter information using Windows 11 Settings, the Control Panel, Command Prompt (CMD), and PowerShell.
- Use **ipconfig**, **getmac**, **hostname**, **ping**, **netstat**, **route print**, and **systeminfo** in CMD, and **Get-NetAdapter**, **Get-NetIPAddress**, and **Test-Connection** in PowerShell.
- Interpret a sample **ipconfig /all** output from a networked computer: identify its IP address, subnet mask, default gateway, and DNS servers, and classify the address as public or private.
- Relate each tool and finding to the layer of the OSI model where it works.
- Design a simple network and IP address plan for the computer laboratory.

II. Materials Needed

- A computer running Windows 11 (a network or Internet connection is **not** required for this activity)
- This worksheet (printed copy, or the digital file to fill out and submit)
- Week 6 lecture notes, for reference on IP addresses, protocols, the OSI model, and network devices

III. Background

A computer does not need to be connected to a network to have network settings. Every PC has one or more **network adapters** (NICs), each with a permanent **MAC address**, and Windows always runs the TCP/IP software, including the **loopback** address 127.0.0.1 (IPv4) and ::1 (IPv6) that the computer uses to talk to itself. Windows 11 lets you inspect these settings in four places: the **Settings** app, the classic **Control Panel**, **Command Prompt (CMD)**, and **PowerShell**. In this activity, you will explore all four on your own unit, observe how Windows behaves when no network is available, and then analyze a sample configuration taken from a computer that *is* connected to a network.

IV. Procedure

Reminder: This is a *viewing* activity. Do not change, disable, or reset any network setting. Close every dialog with **Cancel**. Record values exactly as Windows shows them.

Part A – Windows 11 Settings

1. Press **Windows + I** to open Settings, then click **Network & internet**. Write down the connection status shown at the top of the page (for example, "Not connected").
2. Click **Advanced network settings**. Under **Network adapters**, note every adapter listed (for

example, Ethernet, Wi-Fi, Bluetooth Network Connection) and whether each one is enabled or disconnected.

3. Scroll down and click **Hardware and connection properties**. For each adapter, record its **Name**, **Description**, **Physical address (MAC)**, and **Status** in **Table 1**.

Part B – Control Panel (Network Connections)

4. Press **Windows + R**, type **ncpa.cpl**, and press **Enter**. This opens the classic **Network Connections** window. A red X on an icon means the adapter has no network connection.
5. Right-click your **Ethernet** adapter and choose **Properties**. List the items checked under "This connection uses the following items" in **Table 2**.
6. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**. Record whether the computer is set to **Obtain an IP address automatically** (DHCP) or **Use the following IP address** (static), and the same for DNS. Click **Cancel**, then **Cancel** again. Do not change anything.

Part C – Command Prompt (CMD)

7. Press the **Windows key**, type **cmd**, and press **Enter**. Run each command below, one at a time, and record the results in **Table 3**.

   hostname

   getmac /v

   ipconfig /all

   Tip: For a disconnected adapter, **ipconfig /all** shows "Media disconnected" and no IPv4 address, but it still shows the Description and Physical Address.

8. Test the computer's own TCP/IP software using the loopback address. Record whether you got replies and the time. For **ping localhost**, note the address that replied.

   ping 127.0.0.1

   ping localhost

9. Now observe what happens when there is no network. Record the **exact** message each command displays.

   ping 8.8.8.8

   nslookup google.com

10. View the ports your computer is listening on and its routing table. In Table 3, record three LISTENING local addresses with their port numbers, and the Netmask shown for the Network Destination 127.0.0.0.

    netstat -an | findstr LISTENING

    route print -4

11. Run the command below. It may take a few seconds. Scroll to **Network Card(s)** and record how many network cards are installed and the status of each.

    systeminfo

Part D – PowerShell

12. Right-click the **Start** button and choose **Terminal** (or press the **Windows key**, type **PowerShell**, and press **Enter**). Run each command below and record the results in **Table 4**.

    Get-NetAdapter

    Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength

    Test-Connection 127.0.0.1 -Count 4

Get-NetTCPConnection -State Listen | Select-Object -First 5 LocalAddress, LocalPort

Tip: PowerShell commands follow a Verb-Noun pattern (Get-NetAdapter = "get the network adapters"). Compare each result with the matching CMD command from Part C.

Part E – Analyze a Networked Computer

13. The output below was taken from a laboratory computer that **is** connected to a network. Study it, then complete **Table 5**.

Sample Output 1. ipconfig /all (excerpt) from a networked PC

```
Host Name . . . :                 CICT-LAB-PC07
Ethernet adapter Ethernet:

Description . . . :               Realtek PCIe GbE Family Controller
Physical Address . . . :          3C-52-82-A1-7F-09
DHCP Enabled . . . :              Yes
Link-local IPv6 Address . . . :   fe80::8c2d:4a1f:93b2:61e5%12(Preferred)
IPv4 Address . . . :              192.168.10.37(Preferred)
Subnet Mask . . . :               255.255.255.0
Lease Obtained . . . :            Wednesday, October 14, 2026 8:02:15 AM
Lease Expires . . . :             Thursday, October 15, 2026 8:02:15 AM
Default Gateway . . . :           192.168.10.1
DHCP Server . . . :               192.168.10.1
DNS Servers . . . :               192.168.10.1, 8.8.8.8
```

Part F – Design a Network for Our Laboratory

14. Our laboratory does not yet have a network. In the box after Table 5, draw a simple **star topology** network for it. Include at least **four PCs**, one **switch**, one **router** (default gateway), one **printer**, and the **ISP / Internet**. Use lines to show connections.

15. Assign each device a private IPv4 address from the network **192.168.10.0/24**, using **192.168.10.1** for the router. Label each device in your drawing and complete the **IP Address Plan** (Table 6).

Part G – Optional / Bonus

16. In CMD, run **type C:\Windows\System32\drivers\etc\hosts** to *view* (not edit) the hosts file. In one or two sentences, explain what this file is for and how it relates to DNS.

V. Data and Findings

Table 1. Network Adapters in Settings (Part A)

Connection status shown on the Network & internet page: Connected

Adapter Name	Description	Physical Address (MAC)	Status
Wi-Fi	**Intel Wi-Fi 6 AX201**	**AA-BB-CC-DD-EE-01**	**Disabled / Disconnected**
Ethernet	**Realtek PCIe GbE Family Controller**	**AA-BB-CC-DD-EE-02**	**Network cable unplugged**
Bluetooth Network Connection	**Bluetooth Device (Personal Area Network)**	**AA-BB-CC-DD-EE-03**	**Disabled**

Adapter Name	Description	Physical Address (MAC)	Status
Sailor	Realtek RTL8822CE 802.11ac PCIe Adapter	5C:3A:45:F8:F4:01	Connected

Table 2. Adapter Properties (Part B)

Items Checked Under "This connection uses the following items"	Item Checked (continued)
Client for Microsoft Networks	**Internet Protocol Version 6 (TCP/IPv6)**
File and Printer Sharing for Microsoft Networks	**QoS Packet Scheduler**
Internet Protocol Version 4 (TCP/IPv4)	**—**

TCP/IPv4 Setting	Automatic or Static?
IP address	**Obtain automatically (DHCP)**
DNS server address	**Obtain automatically**

Table 3. Command Prompt Results (Part C)

Command / Item	Your Findings	OSI Layer
hostname	**STUDENT-LAPTOP-01**	Application / Layer 7
getmac /v: MAC address and Transport Name	Wi-Fi: AA-BB-CC-DD-EE-01 \ Device\Tcpip\1 Ethernet: AA-BB-CC-DD-EE-02 \ Device\Tcpip\2	Data Link / Layer 2
ipconfig /all: Media State of Ethernet	Media disconnected	—
ipconfig /all: DHCP Enabled	Yes	—
ping 127.0.0.1: replies and time	4 packets received, 0% loss, time <1ms	Network / Layer 3
ping localhost: address that replied	127.0.0.1	—
ping 8.8.8.8: exact message	Destination host unreachable	Network /

Command / Item	Your Findings	OSI Layer
		Layer 3
nslookup google.com: exact message	DNS request timed out; no servers available	Application / Layer 7
netstat -an: three LISTENING addresses:ports	127.0.0.1:135, [::1]:443, 0.0.0.0:5000	Transport / Layer 4
route print -4: Netmask for 127.0.0.0	255.0.0.0	—
systeminfo: number of NICs and status	3 adapters; 2 disconnected/disabled, 1 loopback active	—

Table 4. PowerShell Results (Part D)

Command	What It Showed	Matches CMD? (Y/N)
Get-NetAdapter: Name, Status, MacAddress, LinkSpeed	Lists Wi-Fi, Ethernet, Bluetooth with matching MACs; 0 Mbps for disconnected adapters	Y
Get-NetIPAddress: IPv4 addresses and PrefixLength	127.0.0.1 /8; no other addresses assigned	Y
Test-Connection 127.0.0.1: result	4 replies, 0% loss, latency <1ms	Y
Get-NetTCPConnection -State Listen: two LocalAddress:LocalPort pairs	127.0.0.1:135, 0.0.0.0:5000	Y

Table 5. Sample Output Analysis (Part E)

Question	Your Answer
IPv4 address of PC07	**192.168.10.37**
Subnet mask (and its /prefix)	**255.255.255.0** — **/24**
Default gateway, and what device it usually is	**192.168.10.1** — a **router**
DNS servers	**192.168.10.1**, **8.8.8.8**
Was the address assigned by DHCP or set manually? How do you know?	**DHCP** — "DHCP Enabled = Yes", plus Lease Obtained/Lease Expires timestamps are present

Question	Your Answer
Public or private? Which RFC 1918 range?	**Private** — falls within **192.168.0.0–192.168.255.255** range
IPv4 address of PC07 in binary (four octets)	11000000 . 10101000 . 00001010 . 00100101
MAC address, and the OSI layer where it is used	**3C-52-82-A1-7F-09** — **Data Link Layer / Layer 2**
How long is the DHCP lease?	**24 hours** (from 8:02 AM 14 Oct → 8:02 AM 15 Oct)
If the DHCP server failed, what kind of address would PC07 assign itself?	**APIPA** — Automatic Private IP Addressing: address in **169.254.0.0/16**

Figure 1. Proposed Laboratory Network Diagram (Part F)

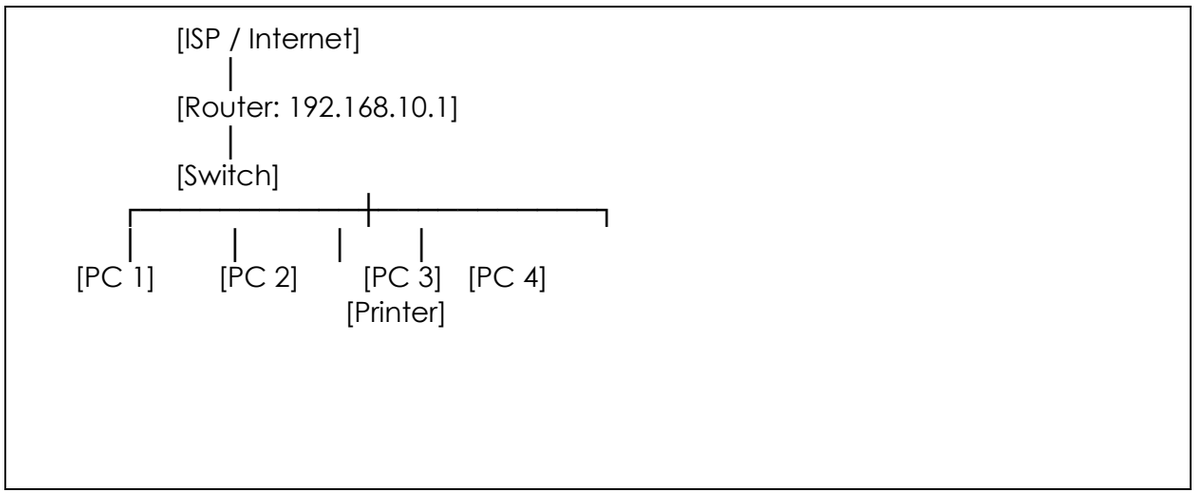


Table 6. IP Address Plan (Part F)

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (gateway)	192.168.10.1	255.255.255.0	—
PC 1	192.168.10.10	255.255.255.0	192.168.10.1
PC 2	192.168.10.11	255.255.255.0	192.168.10.1
PC 3	192.168.10.12	255.255.255.0	192.168.10.1
PC 4	192.168.10.13	255.255.255.0	192.168.10.1
Printer	192.168.10.20	255.255.255.0	192.168.10.1

Part G Notes (Optional / Bonus)

The host's file is a local plain-text file that maps hostnames to IP addresses, checked before sending queries to DNS servers. It serves as a local override: if an entry exists there, the system uses that directly instead of querying DNS, making it the first step in name resolution.

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Even without a network connection, what network information did your computer still have? Why do you think it still has it?

My computer still showed adapter names, MAC addresses, the loopback address 127.0.0.1, and routing for the loopback network. It keeps these because the TCP/IP stack and adapter hardware exist regardless of external connectivity—loopback lets the system communicate with itself for internal services, and MAC addresses are permanently burned into each network adapter.

2. Compare Settings, Control Panel, CMD, and PowerShell. Which tool did you find most useful for viewing network information, and why?

PowerShell was the most useful because it returns structured, consistent results across commands and offers clear, detailed output in a readable format. CMD is powerful but text-heavy and harder to parse; Settings and Control Panel are user-friendly but show less information. PowerShell combines the completeness of command-line tools with better formatting and filtering.

3. What did the messages from ping 8.8.8.8 and nslookup google.com tell you? How would you use messages like these when troubleshooting a real network?

"Destination host unreachable" means there is no path to Google's public DNS—my computer has no active network route. The nslookup timeout confirms DNS resolution is failing, likely because there is no connection to send queries. In troubleshooting, these messages help isolate problems: ping failures point to layer 2/3 connectivity issues, while nslookup failures point to DNS configuration or reachability problems.

4. Why should you be careful about sharing your MAC address, IP address, or ipconfig /all screenshots online? Relate your answer to data privacy.

These details reveal your network setup, hardware identifiers, and location clues. A MAC address uniquely identifies your device and can be tracked across networks; a private IP exposes your internal addressing scheme, and a public IP can pinpoint your location and ISP. Sharing them could let attackers profile you, target your network, or impersonate your device—so treating them as sensitive personal data protects your privacy and security.

VII. Scoring Guide

Criteria	Description	Points
Completeness	All tables (1–6) and required fields are filled out.	25

Criteria	Description	Points
Accuracy	Recorded values match what the tools show and are consistent across Settings, CMD, and PowerShell; sample analysis, binary conversion, and OSI layers are correct.	30
Network Design	Diagram shows a star topology with PCs, switch, router, printer, and ISP; IP plan uses valid, non-duplicate addresses in 192.168.10.0/24.	15
Realization / Reflection	Answers are thoughtful, based on the student's own results, and well written.	20
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	10
Total		**100**